

Calculus I

Continuity

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2019

Outline

1 Continuity

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2 Intermediate Value Theorem

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Continuity

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Definition (Continuous at a Number)

We say that f is continuous at a if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Definition (Discontinuous at a Number)

Suppose f is defined at a . We say f is discontinuous at a if it is not continuous at a .

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Physical phenomena are often continuous. The majority of the physical phenomena that are understood are continuous. Examples:

Discontinuous phenomena examples:

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- Motion of a vehicle with respect to time without sudden brakes.
- Orbits of planets and celestial bodies with respect to time.
- A person's height with respect to time.
- And many more.

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- A person's height with respect to time.
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Discontinuous phenomena examples:

- Particle velocities during collisions and explosions.
- Electric current phenomena, gating events in porins (the event of a molecule passing in and out of a cell).
- Particle physics phenomena.
- And many more.

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- $\lim_{x \rightarrow 4} f(x) \neq f(4)$.

Definition (Greatest Integer Function)

The *greatest integer function* $\lfloor x \rfloor$ is defined as the largest integer that is less than or equal to x .

In computer science this function is called the *floor* function.

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Example

Where is this function discontinuous?

$$f(x) = \begin{cases} \frac{x^2-x-2}{x-2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

Example

Where is this function discontinuous?

- $f(2)$
- $\lim_{x \rightarrow 2} f(x)$

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Where is this function discontinuous?

- $f(2)$ is defined ($f(2) = 1$).
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- Discontinuous at 2.
- This is called a removable discontinuity because we can redefine f at one point to make f continuous.

Example

Where is this function discontinuous?

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- This is called an infinite discontinuity.

Example

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- $f(1)$?
- $\lim_{x \rightarrow 1^+} f(x) = ?$.
- $\lim_{x \rightarrow 1^-} f(x) = ?$.
- $\lim_{x \rightarrow 1} f(x)$?

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- $\lim_{x \rightarrow 1^+} f(x) = 1$.
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- Discontinuous at 1.

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- $\lim_{x \rightarrow 1^+} f(x) = 1$.
- $\lim_{x \rightarrow 1^-} f(x) = 0$.
- $\lim_{x \rightarrow 1} f(x)$ doesn't exist.
- Discontinuous at 1.
- Discontinuous at every integer n .

Example

Where is this function discontinuous?

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- $f(1)$ exists ($f(1) = 1$).
- $\lim_{x \rightarrow 1^+} f(x) = 1$.
- $\lim_{x \rightarrow 1^-} f(x) = 0$.
- $\lim_{x \rightarrow 1} f(x)$ doesn't exist.
- Discontinuous at 1.
- Discontinuous at every integer n .
- The left and right limits both exist but are not equal.

Example

Where is this function discontinuous?

$$f(x) = \lfloor x \rfloor$$

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- $\lim_{x \rightarrow 1} f(x)$ doesn't exist.
- Discontinuous at 1.
- Discontinuous at every integer n .
- The left and right limits both exist but are not equal.
- Such discontinuities are called jump discontinuities (the function appears to “jump”).

Definition (Continuous from the Right or Left)

A function f is continuous from the right at a number a if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

and f is continuous from the left at a if

$$\lim_{x \rightarrow a^-} f(x) = f(a).$$

Example

Consider $f(x) = \lfloor x \rfloor$, and pick any integer n .

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- $\lim_{x \rightarrow n^+} f(x) = ?$.
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Consider $f(x) = \lfloor x \rfloor$, and pick any integer n .

- $f(n) = n$.
- $\lim_{x \rightarrow n^+} f(x) = ?$.
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Consider $f(x) = \lfloor x \rfloor$, and pick any integer n .

- $f(n) = n$.
- $\lim_{x \rightarrow n^+} f(x) = n$.
- Continuous from the right at n .
- $\lim_{x \rightarrow n^-} f(x) = ?$.

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Consider $f(x) = \lfloor x \rfloor$, and pick any integer n .

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Example

Consider $f(x) = \lfloor x \rfloor$, and pick any integer n .

- $f(n) = n$.
- $\lim_{x \rightarrow n^+} f(x) = n$.
- Continuous from the right at n .
- $\lim_{x \rightarrow n^-} f(x) = n - 1$.
- Discontinuous from the left at n .

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- Think of a function that is continuous on an interval as a function that has no breaks in its graph, and so can be drawn “without lifting your pen”.

Theorem (Algebra of Continuous Functions)

If f and g are continuous at a and c is a constant, then the following functions are also continuous at a :

① $f + g$

③ cf

⑤ $\frac{f}{g}$ if $g(a) \neq 0$.

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This shows $f + g$ is continuous at a . The other parts are similar. $\square$

Theorem (Classes of Continuous Functions)

The following types of functions are continuous at every number in their domains:

polynomials rational functions
root functions trigonometric functions

Theorem (Compositions of Continuous Functions)

If g is continuous at a and f is continuous at $g(a)$, then the composition function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at a .

Example

Find $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$.

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- These functions are continuous on their domains, so F is continuous on its domain.
- Its domain is given by $x \neq 3$ and $x \neq -3$.
- Therefore F is continuous on $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$.

Theorem (The Intermediate Value Theorem)

Suppose f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

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- $f(2) = 12$.
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